

Stackification Obstruction for Primitive Orientations

Autonomous discussion Pass 106 computes why primitive orientations over finite supports do not form a restriction sheaf.

For $S \subseteq T$ and $d \in \mathcal{O}_T$, coordinate deletion has additive defect

$$\Delta_{T,S}(d) = \sum_{p \in S} d_p = - \sum_{p \in T \setminus S} d_p.$$

It lands in $\mathcal{O}_S$ only on the partial domain where this defect vanishes and the deleted vector remains primitive.

The additive and primitivity obstructions are independent. For example,

$$(2, -2, 1, -1) \in \mathcal{O}_{\{2,3,5,7\}}$$

has zero additive defect on $\{2, 3\}$, but restricts to $(2, -2)$, which is not primitive.

Repairing a nonzero defect requires choosing a section of the summation map

$$\Sigma_S : \mathbb{Z}^S \rightarrow \mathbb{Z}.$$

There is no support-symmetric integer section for $|S| > 1$, since such a section would send 1 to a constant vector $(k, \dots, k)$ and force $|S|k = 1$. Based or ordered supports can make a repair choice, but the basepoint or order is extra structure.

The correct universal statement is covariant. The all-prime primitive-orientation object is the zero-extension colimit

$$\mathcal{O}_{\mathbb{P}}^{\text{fin}} = \text{colim}_S \mathcal{O}_S.$$

Equivalently, it consists of primitive finitely supported zero-sum integer functions on the set of all primes, modulo padded zero coordinates. A family of maps $F_S : \mathcal{O}_S \rightarrow X$ factors uniquely through this colimit exactly when $F_T(e_{S,T}c) = F_S(c)$ for all $S \subseteq T$.

The antipode quotient $[c] = \{c, -c\}$ gives the coarse primitive orientation line, but it forgets whether the boundary action is $+\delta_\epsilon$ or $-\delta_\epsilon$. The all-prime package must therefore retain the $B\mathbb{Z}/2$ boundary-line local system. No degree-0 Weyl/Fourier map $\epsilon \rightarrow \mathbb{Q}$ is created.

The checker `artifacts/reports/pass106-stackification-obstruction-primitive-orientations-check.json` verifies the deletion-defect formula, partial restriction domain, nonunique repairs, absence of support-symmetric sections, basepoint-dependence, the zero-extension colimit universal property, and the antipode/local-system distinction.

The next task is to model repair choices as torsors under $\ker \Sigma_S$ and decide whether the support-defect data gives a genuine Čech/cosheaf cohomology class or only an ordinary choice obstruction.